

1. Excellence #@REL-EVA-RE@#

1.1 Quality and pertinence of the project's research and innovation objectives (and the extent to which they are ambitious, and go beyond the state of the art)

From photoreceptor signalling to organ morphogenesis: How light regulates leaf shape

1.1.1 Overview:

Leaves perform crucial functions to support plant growth, notably by serving as the primary site of photosynthesis. Photosynthesis depends on the presence of light, and its direction is inferred by detecting the blue part of the light spectrum. In response, plants adjust their morphogenesis to promote light capture. In leaves, blue light from above triggers flattening. By contrast, absence of blue light perception causes downwards curling, and this is enhanced by blue light coming from below. Hence, leaf shape is a plastic trait modified by the surrounding light conditions. Leaf shape also affects gas exchange and transpiration, so has far-reaching effects on the physiology of the whole plant. Therefore, understanding the regulation of leaf shape plasticity is a global scientific priority to facilitate the adaptation of crops to our increasingly variable climatic conditions, and could help to optimise their growth in new artificial light environments. In *Arabidopsis*, blue light induced leaf flattening depends on light perception by the photoreceptors PHOTOTROPIN (PHOT) 1 and 2, and the response is likely mediated by the growth-regulating hormone auxin. From a developmental perspective, organ shape is an emergent outcome of the differential growth of individual interconnected cells, and depends on gene expression domains and tissue-level mechanical properties. It remains unclear how responses at these multiple scales are coordinated to regulate shape, let alone how key environmental cues such as light feed into this regulatory architecture. I propose to answer this question, using the mechanism of blue light-induced leaf flattening as a model. This project will provide a developmental mechanism linking light to organ shape via changes in molecules, cells, and tissues, by accomplishing three objectives:

- 1. Establishing how differential growth patterns drive blue light-induced leaf flattening**
- 2. Identifying how auxin regulates cell growth in leaves in response to blue light**
- 3. Elucidating the key signalling elements of the PHOT photoreceptor in leaves**

This is an interdisciplinary project at the intersection of developmental biology and photobiology, which combines experiments and mathematical modelling to answer a fundamental question about how plants adapt to their surrounding environment.

1.1.2: State of the art and objectives

Leaves grow flat in response to blue light perception from above. This depends on the function of the blue light photoreceptors PHOTs, as evidenced by the constitutively curled leaf phenotype of the *phot1 phot2* mutants^{1,2}. PHOTs are best known for enabling stem phototropism, wherein unidirectional blue light triggers an asymmetric phot activation gradient towards the lit side, causing an inverse auxin gradient, asymmetrical cell expansion, and stem bending towards the light³. In leaves, PHOT expression in the epidermis is sufficient for flattening, and this response occurs after the establishment of leaf polarity⁴. It was proposed that, like in hypocotyls, the response to blue light in leaves depends on a PHOT activation gradient across the organ, namely between the adaxial (top) and abaxial (bottom) leaf epidermis. Developmentally, leaf shape is specified by the differential growth of individual cells across the leaf surface, influenced by gene expression dynamics and tissue mechanics^{5,6,7}. How local changes in growth translate into changes in organ shape is non-intuitive and poorly understood. Here, mathematical modelling has been very useful^{5,8} in helping to propose at least two non-mutually exclusive ways that differential growth causes downward curling^{9,10}: (i) if the top side grows more than the bottom, and (ii) through reduced growth at the edge with tissue asymmetry to bias the buckling downwards (Fig 1A). Therefore, to induce flattening, PHOTs would need to compensate for these effects, by (i) promoting growth at the margin or inhibiting it in the middle, and/or (ii) promoting it on the abaxial epidermis or repressing it on the adaxial one. How PHOT activity affects the spatial dynamics of cell and tissue growth, is not known.

Blue light induced leaf flattening is mediated by auxin, at least in part via regulation of cell expansion. Developing leaves first undergo a period of proliferation, then switch into a cell expansion phase (except for the continued proliferation of the stomatal lineage), before reaching their mature size¹¹. Wild-type leaves curl progressively if they do not perceive blue light from above during the expansion phase⁴, suggesting that PHOTs control flattening at least

¹ Inoue et al. (2008), *Molecular Plant*. 1:15–26

² De Carbonnel et al. (2010), *Plant physiology*. 152: 1391–1405

³ Legris and Boccaccini (2020), *Physiologia Plantarum*. 169:357–68

⁴ Legris et al. (2021), *Plant Physiology*. 187:1235–49

⁵ Bhatia et al. (2021). *Annual Review of Plant Biology*. 2 72:325–56

⁶ Silveira et al (2025), *Nature Plants*. 11:909–923

⁷ Hamant & Saunders (2020), *Annu.Rev.Cell Dev.Biol.* 36:385–410

⁸ Prusinkiewicz and Runions (2012), *New Phytologist*. 193:549–69

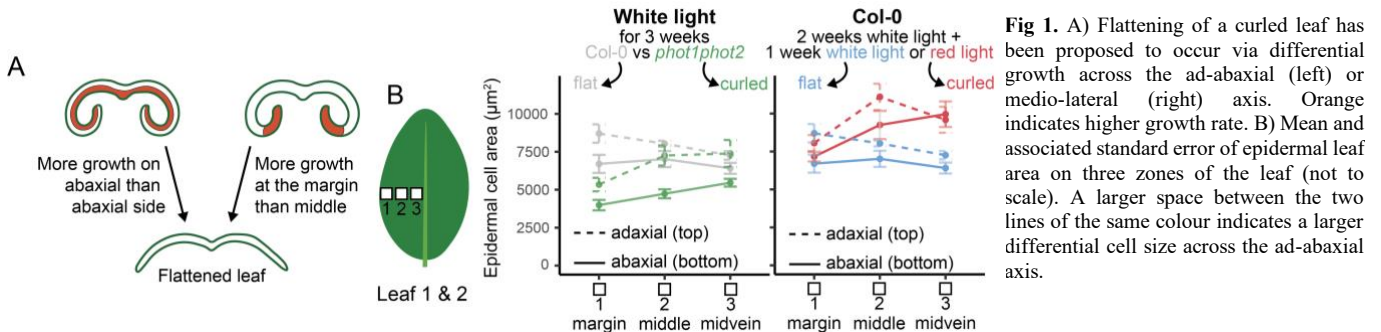
⁹ Coen and Rebocho (2016), *Developmental Cell*. 38:579–583

¹⁰ Boudaoud (2010), *Trends in Plant Sciences*. 15:353–360

¹¹ Maugarny-Calès and Laufs (2018), *Development*. 145:dev161646

partly via cell expansion. This is consistent with the late curling of *phot1 phot2* leaves⁴ and with their reduced cell size¹². *phot1 phot2* leaves also have fewer cells², but the extent of a PHOT-mediated effect on cell division and its contribution to flattening, is not known. Like in hypocotyls, the PHOT response appears mediated by the hormone auxin, consistent with a proposed role promoting cell expansion¹³. Indeed, the localisation of the auxin transcriptional reporter DR5::Venus is sensitive to blue light across the medio-lateral and adaxial-abaxial (hereafter ad-abaxial) axes⁴. Moreover, the auxin transport mutants *aux1 lax1* and *pin3 pin4 pin7* (hereafter *pin347*) are insensitive to blue light induced flattening, with *aux1 lax1* leaves nearly phenocopying those of *phot1 phot2*⁴. Reporters of these genes do have distinct expression across the medio-lateral and ad-abaxial axes in young leaves^{14,15}, but their localizations in older leaves, and whether they respond to blue light in a PHOT-dependent manner, is unknown. If blue light modifies auxin transport patterns across the leaf, this could promote differential cell expansion.

PHOTs are protein kinases located at the plasma membrane that function through interactions with members of the PHOTOTROPIC HYPOCOTYL 3 (NPH3)/ROOT PHOTOTROPISM 2 (RPT2)-like (NLR) and PHYTOCHROME KINASE SUBSTRATE (PKS) protein families. In leaves, PKS3 and NPH3 have been implicated based on defects in blue light induced flattening in their mutants^{1,4}, with *pks3* nearly phenocopying *phot1 phot2*⁴. NPH3 expression was found to be necessary for the response to blue light from above but not from below, hinting at possible ad-abaxial polarity in PHOT signalling⁴. Beyond this, it is not known whether these genes exhibit other spatio-temporal expression patterns across the medio-lateral and/or ab-abaxial axes, how these would influence the flattening response, and how this signalling relates to the putative auxin-regulated growth introduced above.



Data from the host lab suggests that PHOTs may regulate flattening by coordinating growth across the medio-lateral axis, in an adaxial- and abaxial-specific manner. Based on the expected phototropin activation gradient, it is tempting to speculate that blue light induced flattening is regulated solely via differential growth across the ad-abaxial axis (Fig. 1A). In line with this hypothesis, the difference between adaxial and abaxial cell size in *phot1 phot2* leaves is greater than in Col-0 (Fig. 1B, left), Fankhauser lab, unpublished). However, *phot1 phot2* leaves also have a disproportionate reduction in cell size at the margin, consistent with the alternative hypothesis that PHOT induces flattening by promoting growth at the edge of the leaf. Moreover, exposing wild-type leaves to red light during the phase of cell expansion triggers curling and causes clear differential epidermal cell sizes between the margin and centre of the leaf, but not as visibly between the adaxial and abaxial epidermis (Fig 1B, right). **It is not known how, in response to directional blue light, PHOT signalling could cause these spatially restricted changes in growth, and how putative spatial variation in PHOT signalling and auxin signalling/transport would contribute to this response. Such an understanding requires an inquiry into the multi-scale regulation of leaf morphogenesis, aided by mathematical modelling to explore how organ shape depends on emergent properties at cellular and molecular levels.** Nonetheless, it is intriguing that PHOTs may regulate differential growth across the medio-lateral axis, as it suggests that blue light regulates growth across a plane perpendicular to that of its incidence. **Hence elucidating the role of PHOTs in the regulation of leaf flattening also serves as a proof of concept for how the same environmental cue can cause within-tissue spatial variation in signal transduction.**

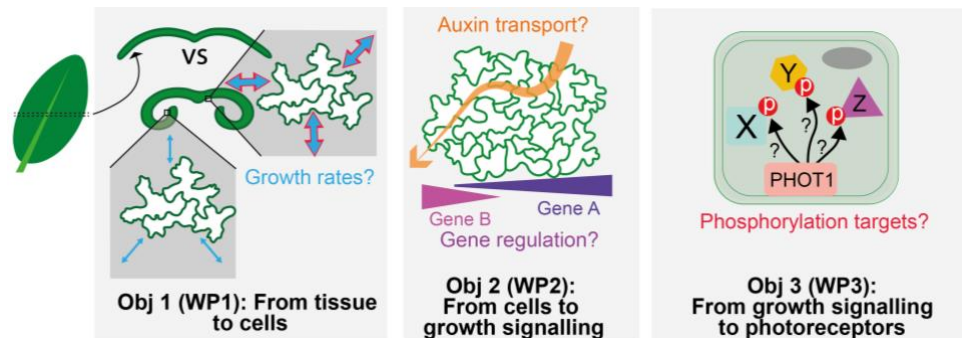


Fig 2. Overview of the three objectives.

¹² Kozuka et al. (2013), *Plant & Cell Physiology*. 54:69-79

¹³ Du et al. (2020), *Annu. Rev. Plant Biol.* 71:379-402

¹⁴ Kasprzewska et al. (2015), *The Plant Journal*. 83:705-18

¹⁵ Verna et al. (2019), *eLife*. 8:e51061

How does directional blue light perception trigger leaf flattening? As a MSCA fellow, I will answer this question, using experiments and mathematical modelling to bridge regulation at key levels of organisation (Fig 2). To achieve this goal, I will pursue three objectives or work packages (WP) that form a coherent project but are designed to be achievable largely independently of each other. Each objective is outlined below, with detailed methodology in [section 1.2](#).

Obj 1 (WP1): Establishing how differential growth patterns drive blue light-induced leaf flattening

I hypothesise that PHOTs induce flattening by promoting growth of the leaf margin in an adaxial- and abaxial-specific manner, such that defective PHOT signalling leads to reduced expansion at the margin and tissue buckling. To test this, sector analysis and live imaging will reveal how blue light from above affects differential growth rates across the leaf. Using these data, I will test whether a cellular mechanical model can recapitulate observed changes in organ shape, in collaboration with the group of Dr Sarah Robinson (University of Cambridge). To understand the relationship between PHOT signalling, differential cell growth, and organ shape, I will generate predictions about the impact of spatially-restricted PHOT1 activation and test these experimentally. Here (and when necessary in subsequent WPs), I focus on PHOT1 rather than PHOT2, because although they both likely play a role, flattening appears predominantly dominated by PHOT1⁴.

Obj2 (WP2): Identifying how auxin regulates cell growth in leaves in response to blue light

The lack of blue light induced flattening in the auxin transport mutant *aux1 lax1*⁴ and in the auxin signalling mutant *arf7 arf19* (unpublished, Fankhauser lab), as well as their near-phenocopy of *phot1 phot2* leaves, suggests that auxin transport and/or signalling are likely involved in this response. In parallel, the effect of PHOT expression on cell size varies across the medio-lateral and ad-abaxial axes (Fig. 1B). This could be due to non-cell autonomous regulation, consistent with a role for auxin transport, and/or to variation in the transcriptional response to auxin, consistent with the restricted patterns of DR5 signal observed in response to blue light⁴. I hypothesise that blue light triggers spatial gradients of auxin transcriptional response and a redistribution of auxin transport, which together modify the patterns of cell growth in the leaf. To test this hypothesis, I will use spatial transcriptomics, live imaging of auxin transporters, and mathematical modelling.

Obj3 (WP3): Elucidating the key signalling elements of the PHOT photoreceptor in leaves

I hypothesise that the impact of PHOT expression on cell size varies across the ad-abaxial and medio-lateral axes (Fig 1B) at least in part due to spatial variation in the phototropin signalling pathway across these axes. PHOTs may cause differential cell growth by phosphorylating spatially-restricted auxin transporters, thereby modulating their activity and hence auxin distribution across the leaf. I will test this hypothesis using live imaging of known phototropin interacting partners, phosphoproteomics to identify new adaxial- and abaxial-specific PHOT targets, and biochemical approaches.

Innovative aspects: This project will provide a mechanistic account of how responses at successive levels (intracellular signalling, growth signalling, tissue growth, organ shape) are coordinated in response to light, enabling the adaptation of leaves to light conditions. This represents a significant advance for the fields of photobiology and development. The response of leaves to light is crucial for plants to optimise light capture, yet photoreceptor signalling pathways, including phototropins, have mostly been studied in hypocotyls. The project will address this gap, which also provides the opportunity to explore questions that are not accessible in hypocotyls, such as how organ polarity (e.g. ad-abaxial) modifies photoreceptor signalling. Moreover, the model, supported by experiments, will provide a predictive understanding of how genetic and cellular dynamics are translated into changes in organ shape, a fundamental yet poorly understood question in developmental biology. The ambition and innovativeness of this work stems from its integration of developmental biology and photobiology, and is made possible by the synergy between the Fankhauser lab (photobiology, biochemical techniques, -omics), my own expertise (plant development, auxin regulation, combining modelling with experiments), our collaboration with the group of Dr Sarah Robinson at the University of Cambridge (leaf development and mechanics, including modelling), and the state-of-the-art facilities at UNIL (see [1.3](#) & [3.2](#)). Therefore, this project would not be possible at another institute. Furthermore, the medio-lateral variation in the effect of PHOT on cell size (Fig 1B) represents a case study for how cells of the same tissue exhibit a spatially defined response to environmental signals. These questions are only now becoming accessible thanks to the development of single-cell spatial transcriptomic technologies for plants. In enabling a high-throughput visualisation of gene expression, this technology allows for unprecedented insights into the within-tissue spatial regulation of gene expression.

1.2 *Soundness of the proposed methodology (including interdisciplinary approaches, consideration of the gender dimension and other diversity aspects if relevant for the research project, and the quality of open science practices).*

Leaves 1 and 2 will be used throughout and leaf flattening index will be measured as previously in the Fankhauser group^{2,4}. **All experiments will be done in red light (RL), red light supplemented with blue light (RL+BL), and white light (WL), as previously⁴.** Given I focus on blue light which suffices to flatten leaves, R+BL treatments is a more specific treatment, but I also use WL as a control. A consideration for several WPs (confocal imaging and spatial transcriptomics) is the large size of leaves at late developmental stages. **In those cases, I will focus on the band of tissue in the middle of the leaf stretching from the midvein to the margin, where cells are known to respond to blue light in a PHOT-dependent manner (fig 1B).** So far, observations in this region have been representative of those throughout the medio-lateral axis, though other regions will be inspected and analysed as/if needed, depending on the findings that emerge. Using a Stellaris confocal microscope will enable automated imaging of tiles (including adjustments in Z plane) and stitching, which makes the imaging goals below achievable.

WP1) Establishing how differential growth patterns drive blue light-induced leaf flattening

Task 1.1: Characterise where, when, and how PHOTs affect leaf growth rate. I will perform clonal sector analysis using an existing Col-0 line with constitutively driven (35S promoter) Heatshock-inducible CRE-mediated lox site recombination that induces GFP expression¹⁶, which the lab is already crossing into *phot1 phot2*, *nph3* and *pks3*. *nph3* and *pks3* are PHOT interactors with blue-light flattening defects. A short (~3 min) heatshock will induce GFP expression in several cells (sectors) randomly across the leaf of plants. As the leaf continues to grow post-treatment, the GFP sectors become larger as each induced cell expands and/or divides. Therefore, sector size can be quantified to infer growth rate, with smaller sectors indicating slower growth. I will compare sector size across the medio-lateral and ad-abaxial axes 3 days post-induction when the lines above are grown in RL, WL and R+BL. This will reveal how PHOT signalling and its interactors modify differential growth rates across the two axes. Induction at several time points during the proliferation and expansion phases (6, 9, 14 days) will reveal when in development the differential growth rates are most prominent. Although the GFP visualisation wavelength also activates PHOT1, this is not a significant limitation given I am imaging at the end of the experiment. Moreover, comparing the number of cells in sectors will indicate whether PHOT signalling mainly regulates cell expansion, as hypothesised, or whether it also modifies cell division rates. Sectors will be segmented with CellPose. Sector analysis is a classical and powerful method in plant development, but that has not been used by photobiologists. It enables growth rates to be tracked in large tissues and on the adaxial leaf epidermis (otherwise difficult because of trichomes), and is therefore the ideal tool to study leaf flattening. It is also used by the Robinson group who will provide support if needed, although the methodology is reasonably straightforward.

Task 1.2: Test whether a mechanical model can recapitulate the observed leaf shapes. This will utilise a leaf-shaped cellular bilayer mesh model wherein each cell has a customisable growth rate. The model is already being developed from existing software¹⁷ and for other purposes, by Dr Euan Smithers an expert computational modeller in the lab of Dr Robinson. I will draw on my own modelling experience to engage with its technical aspects, and once the model developed, use it to generate predictions. The model abstracts the inside of the leaf, consistent with epidermal PHOT1 expression rescuing the *phot1 phot2* leaf phenotype⁴. We will input the experimentally observed growth rates and test whether those can generate the observed changes in shape. Discrepancies between model and experiments will generate hypotheses about the presence of additional feedbacks between cells growth and tissue shape, which will be explored using the model.

Task 1.3: Generate and test model predictions. Once the model developed, I will generate model predictions of the change in leaf shape if only the adaxial or abaxial side responds to blue light, or only part of the medio-lateral axis. Because shape is an emergent property of the model, it is difficult to anticipate what the model will predict, which underscores its utility. Simulations will be compared to the shape of leaves in existing lines with adaxial- or abaxial-driven PHOT1 expression in *phot1 phot2* (under pAS2 and pFIL), which are known to fully and partially rescue the *phot1 phot2* phenotype, respectively⁴. To selectively induce PHOT1 expression across part of the medio-lateral axis, transgenic lines will be generated carrying a Cre/Lox inducible PHOT1:GFP driven by the adaxial AS2 or abaxial FIL promoters in *phot1 phot2*. I will use these to generate leaves with large sectors of PHOT1 expression induced across only part of the medio-lateral axis and compare flatness and cell growth of induced and non-induced

¹⁶ Fitzsimons et al. (2025), *BioRxiv*

¹⁷ Bonfanti et al. (2023), *PNAS*. 120: e2302985120

sections to model predictions (staining with calcofluor white for non-induced sections). If the induced sectors rescue leaf flattening better than predicted by the model, this could point to other feedbacks, such as non-cell-autonomous signalling, which will be explored further in WP2.

WP2) Identifying how auxin regulates cell growth in leaves in response to blue light

Task 2.1: Perform single-cell spatial transcriptomics of leaves. I hypothesise that the medio-lateral gradient in cell size in curled leaves (Fig 1B) is due to yet-unknown PHOT-mediated cell-specific expression patterns that influence the auxin response. I will perform single-nuclei RNAseq (the method of choice for aerial plant tissue) on WL, RL, and R+BL-grown Col-0 and *phot1 phot2* to uncover genes expressed in a PHOT-dependent manner under blue light. I will use a reporter line driven by the epidermal ML1 promoter, followed by fluorescence-activated cell sorting (FACS) to enrich for epidermal cells. A panel of 500 genes will be generated based on differential expression and known or hypothesised involvements in leaf growth, auxin regulation and/or phototropin signalling. These will be used for multiplexed error robust fluorescence in situ hybridization (MERFISH) (Vizgen), excluding highly expressed genes as they interfere with MERSCOPE imaging. MERFISH is an innovative technology to profile single-cell gene expression across tissues, and has successfully been used in Arabidopsis leaves¹⁸ (I am currently trying to obtain access to existing data which will also help generate preliminary hypotheses). The MERSCOPE platform at UNIL enables imaging of up to 1cm² tissue sections, large enough for the region of interest (midvein to margin). New candidate genes will be included in subsequent work as appropriate. The time points chosen will depend on when I observe the largest differential growth rates in task 1.1, and will likely involve at least two.

Task 2.2: Image auxin transporters and transcriptional reporter. I will investigate whether blue light modifies auxin transport in a PHOT-dependent manner, focusing on reporters for AUX1, LAX1, and PIN347, given *aux1lax1* and *pin347* leaves are insensitive to blue-light induced flattening⁴. Confocal imaging will be performed on existing reporter lines to assess changes in medio-lateral and/or ad-abaxial expression, and varying light and *phot1 phot2* background. Transverse cuts will also reveal changes in orientation of PIN polarity. I will follow the same steps to image the qDII reporter for auxin transcriptional input, which has been used as a proxy for auxin distribution¹⁹.

Task 2.3: Incorporate results on changes in auxin transport and signalling into the mathematical model. This builds on the model developed in task 1.2. Using a simple compartmental ODE system, we will simulate how the putative regulators of auxin signalling and/or transport identified above modify cell expansion patterns in different light conditions and mutants, assuming cell expansion occurs in regions of high auxin. I will test whether these simulations recapitulate the experimentally observed cell expansion patterns using the sector analysis from task 1.1. Discrepancies between model and data will point to alternative testable relationship between auxin transport/signalling, cell growth, and any other regulator identified in task 2.1. The model will generate predictions that I will test via further pharmacological and genetic perturbations of auxin synthesis, signalling, and transport.

WP3) Elucidating the key signalling elements of the PHOT receptor in leaves

Task 3.1: Image reporters of PHOT signalling elements. I will use confocal imaging to test whether blue light modulates expression and localisation of NPH3 and PKS3. These (or other members of the same family) interact with PHOTs in hypocotyls and mutants in these genes have defective blue light induced flattening⁴. The Fankhauser lab has expertise creating knock-in (KI) lines to visualise fluorescent reporters for genes expressed within their genomic environment, and is generating KI lines of PKS3 and NPH3. I will image these reporters, varying light conditions and genetic background (at least *aux1lax1*, *phot1 phot2*, others may emerge from task 2.1), and assess any medio-lateral and/or ad-abaxial expression patterns. Moreover, insights from task 1.1 about how leaf growth rate is affected in these mutants will be useful here. I will compare where each reporter is expressed with where the growth rate is affected in that mutant, which will provide insights into their putative regulation of cell growth.

Task 3.2: Identify and characterise tissue-specific PHOT1 targets. Phosphoproteomics will be done on BL, R+BL, and WL grown leaves of Col-0, *phot1 phot2*, and existing lines of *phot1 phot2* with PHOT1 expression driven across the whole epidermis (ML1::PHOT1), and on the abaxial (FIL::PHOT1) or adaxial side (AS2::PHOT1). The last three lines will enable enrichment for epidermal, abaxial, and adaxial phosphorylation targets. This will reveal whether any identified targets are specified by leaf polarity. I will compare the targets obtained to existing datasets in the Fankhauser lab from etiolated seedlings. This may help select candidates for further analysis, by highlighting proteins, or types thereof, which are common targets across these two contexts. Moreover, putative mechanisms emerging from the other strands of this project may also point to the most promising candidates on which to follow-

¹⁸ Nobori et al. (2025), *Nature*. 638:197-205

¹⁹ Galvan-Ampudia et al (2020), *eLife*. 9:e55832.

up. Subsequent work will involve assessment of mutant phenotypes, reporter expression, as well as the standard biochemical approaches in the Fankhauser lab (immunolocalisation, western blot).

Interdisciplinarity: Interdisciplinarity is integral to the proposed project, as it combines mathematical modelling with a range of classical and cutting-edge experimental techniques (microscopy, genetics, biochemistry, spatial transcriptomics). Mathematical models are powerful tools to generate testable predictions about the relationship between different levels of regulation (molecules, cells, tissues), which often exhibit emergent properties and hence non-intuitive dynamics. By uncovering the regulatory layers linking photoreceptor signalling to organ morphogenesis, this methodology enables the integration of photobiology and developmental biology.

Gender dimension: This aspect is not relevant given this work is performed on the plant *Arabidopsis*.

Open-science practices: To promote early dissemination, articles will be posted on *BioRxiv* first, and then sent for peer-review. To ensure accessibility and reproducibility, manuscript submissions will include all raw data and code underlying data analysis and modelling. These practises are also followed by the Robinson group, and I have already implemented them for my PhD publications. Published papers will be made open-access. Constructs and transgenic line generated will be available upon request. To promote reusability, dataset from the phosphoproteomics and spatial transcriptomics will be uploaded to databases operated by the European Bioinformatics Institute, namely the Proteomics Identifications Database and the Functional Genomics Data Collection (ArrayExpress), respectively.

Research data management: Every member of the Fankhauser lab reports new material generated (plasmids, probes, antibodies, transgenic seeds, etc) in inventories that are in line with FAIR principles. This is supported by the help of the two technicians in the lab who have been managing lab resources for 20 years. I will have an electronic lab book where I clearly document all experiments and protocols. Code for data analysis and modelling will be annotated to ensure reusability.

1.3 *Quality of the supervision, training and of the two-way transfer of knowledge between the researcher and the host*

Qualifications and experience of the supervisor: The project is hosted by Prof. Fankhauser at the Center for Integrative Genomics (CIG), University of Lausanne (UNIL) (see also B2-5.2). For over 20 years, he has been leading research in plant photobiology and development, including via modelling collaborations. He has supervised and mentored many postdocs, including MSCA fellows, towards independent academic positions. In the past 5 years, 4 postdocs became Assistant Professors in France, Switzerland, Denmark, and India. The high quality of science in his lab is evident from the 100+ publications, many in high impact journals (*Science*, *PNAS*, *Nature Comms*). He collaborates internationally with leading scientists (e.g., Jorge Casal, Argentina, Matyáš Fendrych, Czech Republic).

Planned training activities: The numerous training activities afforded by this fellowship will be invaluable in helping me transition to an independent position in academia. Given his experience, Prof. Fankhauser is the ideal mentor to reach this career goal. We will have weekly scientific meetings and generate a career development plan from the onset of the project, which will be reviewed at biyearly career meetings. Scientifically, I wish to develop knowledge and skills in photobiology, molecular biology, protein biochemistry, and -omics analyses. These are areas of expertise in the lab. I will receive technical training from the host lab, and bioinformatic training for the single-cell analysis via the joint courses *Introduction to bulk RNA-seq: from quality control to Pathway Analysis* and *Single-cell transcriptomics with R* offered by the Swiss Institute for Bioinformatics. The single-cell analysis will also benefit from an ongoing collaboration with the lab of Keiko Sugimoto (RIKEN), who has experience in this area. I will also hugely expand my skills by interacting with the many facilities that will support the several experiments in this project (see 3.2 and B2-5.2). Moreover, I will extend my transferrable management and teaching skills by attending the EMBO Lab Leadership for Postdocs, by supervising at least one masters student, and by teaching undergraduate practical courses (e.g. on Experimental Design). My communication skills will be enhanced from giving yearly seminar at the CIG and at the neighbouring plant molecular biology department (DBMV), with which we share seminars. Attending these seminars weekly will also teach me about new research and techniques in both plant biology and other eukaryotic systems. Finally, the UNIL course *How to prepare a SNSF career application* will prepare me (CV writing, budget) to further my scientific independence after my MSCA tenure. The above training will be documented by building a personal portfolio that will be re-assessed yearly.

Transfer from the host to the applicant: The Fankhauser lab members are experts in molecular biology, biochemistry, photobiology, microscopy, and bioinformatics. I will benefit from their support when learning new techniques and troubleshooting experiments. In particular, I will be taught the biochemistry and cloning protocols by

two senior technicians who are highly skilled molecular biologists and biochemists, and who have worked in the lab for over 20 years. I will also get in-depth feedback on my project from all members via lab meeting presentations every 3 months, as well as regular informal feedback and support during daily interactions.

Transfer from the applicant to the host: My areas of expertise (plant development, auxin-mediated regulation, combined use of experiments and mathematical modelling) complement those of the other lab members outlined above. For example, no current lab member has modelling experience. Therefore, my input during lab meetings and daily interactions will enrich the group's skillset, which may foster further innovative interdisciplinary collaborations with other teams.

1.4 *Quality and appropriateness of the researcher's professional experience, competences and skills*

During my academic training, I have developed strong scientific and transferrable skills that I will draw on to ensure the success of my MSCA fellowship. I became fascinated by plant development during my BA in Biological Sciences at the University of Oxford, where I complete a research project in the group of Prof. Liam Dolan. I then undertook my PhD at the Sainsbury Laboratory University of Cambridge (SLCU), a leading institute for interdisciplinary and quantitative approaches to plant development. I was supervised by Prof. Ottoline Leyser, a world-renowned expert in plant development and auxin signalling. During this time, I drove an interdisciplinary project on the regulation of shoot branching regulation in *Arabidopsis*, focusing on auxin transport and its self-organising properties. I learnt several skills relevant to the proposed project, including how to work with *Arabidopsis*, confocal microscopy techniques, and programming in R and Julia (see also B2-CV). I developed notable expertise in (i) the auxin regulation of plant development and (ii) the use of experiments and modelling to generate hypotheses and testable predictions. I believe that bringing this expertise to the proposed project will enable a step-change in our understanding of how light influences leaf shape, a question which so far has largely been investigated by photobiologists.

My PhD work led to two first-author publications^{20,21}, including one as co-corresponding author, in reputable plant science and broad life science journals (*New Phytologist*, *PLOS Biology*). I am proud to have developed, managed, and completed this PhD work with considerable independence. Six months after I started, Prof. Leyser took on the leading role of the largest UK research funding agency (UKRI). While I continued to receive guidance from her on the scientific direction of my project, this transition led to the progressive closure of the lab. I capitalised on this opportunity to create my own scientific trajectory. I joined the group of Prof. Locke to build a mathematical model that would test the hypotheses obtained from my experimental results, and learnt dynamical systems modelling from a senior PhD student in the group. He led the initial model development, which I soon pushed forward independently. Through this experience, I learnt to seek out the new mentors needed to push forward innovative research questions. During my PhD, I also developed a strong international network and excellent communication skills by giving talks at conferences and seminars, including as an invited speaker. I acquired transferrable management skills through organisational responsibilities (e.g. organiser for the interdisciplinary Cambridge Theory of Living Matter seminars over 3 years). I taught undergraduates for 2 years (setting and marking essays, preparing discussions), where I learnt how to adapt my pedagogy to the needs of individual students.

2. **Impact #@IMP-ACT-IA@#**

2.1 *Credibility of the measures to enhance the career perspectives and employability of the researcher and contribution to his/her skills development*

I wish to pursue a career in academia, leading a research group on plant developmental plasticity. I am particularly interested in how organisms integrate complex environmental information and how developmental plasticity arises from orchestrated responses across level of organisation. Investigating the relationship between the genetic, protein, cellular, and tissues scales, is inherently a multidisciplinary endeavour. Therefore, achieving my career goal requires a broad and interdisciplinary skillset, which will be tremendously enriched through the training opportunities afforded by this fellowship. Within the lab, I will learn protein biochemistry, cloning, and -omics technologies. I will also advance my microscopy skills thanks to the expertise of other renowned groups working at the DBMV on *Arabidopsis* (Niko Geldner, Christian Hardtke), and at the CIG (Nadine Vastenhouw). The project will also benefit from the mechanics expertise of Prof. Mateusz Majda at the DBMV. The collaboration with Dr Sarah Robinson (University of Cambridge) and Dr Sugimoto (RIKEN), will expand my skills in modelling and single-cell analyses, respectively. Given the innovativeness that will come from our combination of skillsets, it is likely that the project

²⁰ Nahas et al. (2024), *New Phytologist*. 242:1084–1097

²¹ Nahas et al. (2025), *PLOS Biology*. (accepted)

will create durable collaborations that will enrich the rest of my scientific career. My scientific network will also grow thanks to the local and international leaders, including in plant biology, who regularly give seminars at the DBMV and CIG. Together with the opportunities to develop my transferrable skills (management, leadership, teaching, see [1.3](#)), these activities will greatly enhance the competitiveness of my academic profile, helping me establish further independence after my MSCA tenure. I will apply for Swiss National Science Foundation (SNSF) Ambizione and Starting grants during the second year of this fellowship. Should I not secure an academic position, the skillsets acquired will make me highly employable for senior roles in industry. I will join a leading European vegetable-breeding company, where I will apply my scientific and transferrable skills to generate more resilient crops.

2.2 *Suitability and quality of the measures to maximise expected outcomes and impacts, as set out in the dissemination and exploitation plan, including communication activities #@COM-DIS-VIS-CDV@#*

Dissemination and exploitation: Papers generated will be posted on *BioRxiv* with all raw data and scripts. Papers and ongoing news will be disseminated on bluesky (zoe.nahas@bsky.social) where I run an active account with 700+ followers (mainly researchers from across the life sciences) and will write a thread to accompany each publication. During the fellowship, I will attend at least two international conferences (International Symposium in Plant Photobiology 2026 and FASEB mechanisms in plant development 2027) to disseminate ongoing and unpublished work and obtain feedback from the photobiology and developmental biology communities. By leveraging interdisciplinarity and cutting-edge techniques, the project will uncover principles of environmentally regulated morphogenesis that are relevant to researchers across the life sciences. Therefore, I expect it will lead to at least one important publication in a high-impact generalist journal (*Science*, *PNAS*, *Current Biology*) that will be highly cited.

Communication and public engagement: It is critical that the public be engaged in the scientific process. To further my science communication skills, I am joining the European Science Engagement Association, which focuses on fostering public engagement. I will attend their 2026 conference in Luxembourg attended by experienced communicators and educators from across Europe. I will apply the skills learnt by organising an outreach activity on developmental plasticity as part of “Mystère de L’UNIL”, a yearly university-wide event that receives ~10000 attendees, from families and individuals from across the region. I will grow *Arabidopsis* and two well-known edible plants (nasturtium and lettuce) in different light conditions (very high light, in the sun, in red light with or without blue light, all of which influence size and/or shape) in the plots outside the CIG building and in the spare space of the growth chambers. I will exhibit the differences in form between these plants so that people can interact with them, and will explain how the organ shapes visible by eye are due to changes at cellular, genetic, hormonal levels. This will be supported by a poster and simple computer simulations from the project to illustrate how organ shape emerges from differential growth between cells. The event will be shared on the UNIL Instagram account to disseminate it beyond the local community.

Intellectual property: Should results of this research require intellectual property protection, this will be raised with the transfer technology office of Lausanne that is dedicated to these purposes, including by supporting start-ups and product commercialisation.

2.3 *The magnitude and importance of the project’s contribution to the expected scientific, societal and economic impacts*

Scientific impact: The project will uncover principles of morphogenesis applicable to many other plants and plant organs, so the results will be relevant to all members of the developmental biology community. It will be interesting to compare the mechanisms identified with those regulating the morphogenesis of comparatively invariable organs such as petals, where the need for phenotypic robustness has made their development rather insensitive to environmental condition. Comparisons with the mechanisms regulating the shape of other flat animal organs, such as wings or intestines, will also inform the extent to which differences between plants and animals (e.g. in cell mobility) have influenced the evolution of developmental mechanisms. Moreover, given other environmental cues (e.g. shade, drought) modulate leaf flatness, photobiologists, physiologists, and developmental biologists working on these subjects will be able to come together to use and/or expand the model to generate hypotheses about what mechanisms might be operating to mediate these environmental responses, and how multiple environmental cues are integrated simultaneously. Hence the project will create opportunities for more innovative interdisciplinary research.

Economic/technological impact: The knowledge gained could be leveraged to optimise artificial light environments, for example in vertical farms. Beyond agriculture, understanding how leaf shape is modified by light will inform how we breed for different leaf shapes in ornamental plant. Although the proposed study focuses on *Arabidopsis* due to the unmatched availability of resources that enables the undertaking of ambitious research project like the one

proposed, I am confident that the findings will be applicable to other plants including crops. Indeed, Arabidopsis research has successfully provided translational insights by identifying core genes and developmental pathways²², and lettuce leaves have already been shown to be influenced by blue light in the same way as Arabidopsis²³.

#§COM-DIS-VIS-CDV§# #§IMP-ACT-IA§#

3. Quality and Efficiency of the Implementation #@QUA-LIT-QL@# #@WRK-PLA-WP@# #@CON-SOR-CS@# #@PRJ-MGT-PM@#

3.1 Quality and effectiveness of the work plan, assessment of risks and appropriateness of the effort assigned to work packages

Table 1: Gantt chart for the proposed project (when necessary, task name is condensed compared to in section 1.2)

Tasks (T) + Deliverables (D) + Milestones (M)	Months																							
	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24
WP1									D1															
T1.1: Perform sector analysis in leaves																								
T1.2: Reproduce leaf shapes with mathematical model																								
T1.3: Generate and test predictions from the model																								
M1: Predictive spatio-temporal understanding of the relationship between cell growth and leaf shape in response to blue light									M1															
WP2																								
T2.1: Single-cell spatial transcriptomics of leaves																								
T2.2: Imaging auxin reporters and transporters																								
T2.3 Incorporating results into mathematical model and testing model predictions																								
M2: Established the relationship between cell growth, auxin signalling, and auxin transport, in response to blue light																		M2						
WP3																								
T3.1: Imaging of PHOT signalling elements																								
T3.2: Identify and characterise new PHOT1 targets																								
M3: Identified the spatial distribution of core signalling elements linking PHOT1 to auxin and cell growth																			M3					
WP4 - Management	D2																							
T4.1 Biweekly supervision meetings																								
WP5 - Career development and training	D3																							
T5.1 Joint bioinformatic courses																								
T5.2 Career mentoring meetings																								
T5.3 Course on Preparing an SNSF application																								
T5.4 EMBO Leadership workshop																								
T5.5 SNSF Ambizione and starting grant applications																								
WP6 - Dissemination and communication:	D4																							
T6.1 Present at lab meetings and departmental seminars																								
T6.2 Present at two conferences																								
T6.3 European Science Engagement conference																								
T6.4 Mystere de l'UNIL outreach																								
T6.5 Write draft manuscript																								
T6.6 Submission and revision of manuscript																								

Given Arabidopsis has a 6–8-week life cycle, generating transgenic lines and crosses is a long process. Hence the cloning in T1.3 (to generate PHOT1 inducible lines under pFIL and pAS2), and the crosses of reporters in mutant backgrounds (T2.2 and T3.1) will be pushed forward from the start of the project. This is reflected in the early start

²² Yaschenko et al. (2025), *The Plant Cell*. 37:koae065

²³ Wang et al. (2021), *Int J Mol Sci*. 22: 3157

of T1.3 in **Table 1** but is not shown explicitly for crosses given this is less demanding work (though is considered when indicating the timing of T2.2 and T3.1). The key first experiment is the sector analysis (T1.1) to identify when and where the differential growth patterns occur. This experiment is relatively easy, and the crosses are already being generated by the lab, so they will be ready by the start of the fellowship. These results will help to constrain the regions of the leaf and timepoints which should be used for and single-cell transcriptomics/MERFISH and confocal imaging (although more exploratory confocal imaging will also be done). Given the importance of WP1, I will generate a report (D1) on its conclusions and implications for WP2 and 3. Sowing will be staggered appropriately so that plants for different experiments can be grown in parallel. During the periods of modelling, I will meet with Dr Smithers weekly on Microsoft Teams, and fortnightly with both of our supervisors. I will produce a data management plan at the start of the project (D2) using the recommended template, a career development plan will be developed with Prof. Fankhauser (D3) and a communication and dissemination plan (D4). I will submit the manuscript to BioRxiv 4 months before project end (D5), with the aim of having it published by the end of my MSCA tenure (D6).

Table 2: Risk assessment of project and actions to mitigate them. L= Low, M = Medium, H = High

Risk	Likelihood	Consequence	Action to mitigate risk
(WP1+2) Modelling collaborator leaves current position	M	L	I will proactively learn key technical details of the model so I can use it independently once it has been developed. This is feasible given my experience.
(WP2) No changes in expression or localisation of AUX1/LAX and PIN3/4/ reporters despite mutant phenotypes	L	M	May indicate that the mutants are impaired in blue light induced flattening for another reason, e.g. ad-abaxial polarity defect. This will be highly informative given it is poorly understood how this polarity contributes to flattening. I will pre-emptively cross these mutants with ad-abaxial polarity markers from the beginning until the T2 generation, given this is very little work and will enables quick follow-up as/if needed.
(WP3) pFIL/pAS2 expression levels too low for phosphoproteomics	M	L	I will pre-empt this by mining the latest most comprehensive single nucleus RNAseq of the Arabidopsis lifecycle ²⁴ to investigate alternative promoters.
(WP4) Publications not complete by the end of the project	L	M	I am planning to finish all experiments 4 months before the end of fellowship to give myself plenty of time for writing publications and plan my next career steps.

3.2 *Quality and capacity of the host institutions and participating organisations, including hosting arrangements*

UNIL is a major hub for plant research and hence an exceptional place for my postdoctoral training. The CIG, where I will be located, hosts 14 research groups working on a range of eukaryotic systems. This will enable me to get creative inputs from biologists in different fields, while still receiving support from plant scientists thanks to the regular contacts with the DBMV across the road (at least weekly for joint seminars). As part of the Faculty of Biology and Medicine, I will benefit from access to state-of-the-art facilities to support the experiments proposed (see B2-5 for the necessary details). Each facility is run by highly skilled staff providing personalised assistance to design, implement, analyse, and troubleshoot experiments. Moreover, two senior technicians in the Fankhauser group have worked in the lab for over 20 years and are experts in molecular biology and protein biochemistry. They will teach me the necessary cloning and biochemistry protocols. In addition to the 2 technicians, the Fankhauser group consists of 4 postdocs and 3 PhD students. I will receive support from all of them through informal interactions given I have my own bench and desk near their. The group organises weekly coffee breaks and biyearly outings, fostering a friendly and collaborative atmosphere. In addition, the CIG organizes a yearly retreat, weekly seminars, a fortnightly social events, all of which makes it very easy to integrate this vibrant international community. When preparing my next career steps, I will receive support from grant office of the Faculty of Biology and Medicine (FBM). I will benefit from The Faculty of Biology and Medicine will also has a range of career services on CV writing, and Networks (e.g. Bioscience Network Lausanne) which will help me with my next career steps. Overall, I am confident I will receive all the necessary support to achieve the aims set out above and obtain the skills outlined in section 1.3.
 #§CON-SOR-CS§# #§PRJ-MGT-PM§# #§QUA-LIT-QL§# #§WRK-PLA-WP§#

²⁴ Lee et al. (2025), *Nature Plants*. doi: 10.1038/s41477-025-02072-z